Datapitch - Finance

Anton Pühringer - 12414573
Maximilian Trentini - 11916686
Michele Fu Wang - 12347518
Julian Bridi - 12301867
Jonathan Gabriel - 12319994

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Agenda

- **Stock Return Analysis & Modelling (Fama-French)**
  - Data and License
  - Preliminary Data Explanation
  - Planned Methods
  - Research Question & Potential Challenges
- **Default of Credit Card Clients (PD Modelling)**
  - Data and License
  - Preliminary Data Explanation
  - Planned Methods
  - Research Question & Potential Challenges

Data Sources and Licenses

- **Stock Return Analysis and Modelling (Fama-French):**
  - Fama-French daily 3-factor data (2020-01-01 to 2026-10-30)
    Open academic license (Kenneth French Data Library)
  - Stock prices pulled from Yahoo Finance via `quantmod` (2020-01-01 to 2026-10-30)
    Academic research use aligned with Yahoo-Finance Terms and Conditions
- **Default of Credit Card Clients:**
  - Data on Default of Credit Card Clients in Taiwan (UCI ML Repository, published 2009)
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Preliminary Data: Stock Return Analysis

Fama-French Factors and Asset Returns

Core Variables: - **Risk-free Rate (R_f):** U.S. 1-Month T-Bill - **Market Excess Return ($R_m - R_f$)** - **Size Factor (SMB):** Small Minus Big - **Value Factor (HML):** High Minus Low **Target:** - Excess equity returns ($R_i - R_f$)

Platzhalter für Micheles Fama-French Faktor-Plot (z. B. kumulierte Renditen).

Preliminary Data: Credit Card Defaults

UCI Taiwan Credit Card Dataset

Features ($N = 30\,000$): - **Demographics:** `LIMIT_BAL`, `SEX`, `EDUCATION`, `AGE` - **History:** Repayment status (`PAY_0` to `PAY_6`) - **Financials:** Bill statement amounts & previous payments
Target: - `default.payment.next.month` (Binary, ~22% Defaults)

Platzhalter für Micheles Default-Rate / Imbalance Plot.

Research Methodology and Questions

- **Fama-French Stock Return Modelling:**
  - **Question:** How much of the cross-sectional return variation in a selected portfolio consisting of 100 selected stocks can be explained by factor premia?
  - **Method:** OLS Regression and robust regression techniques applied to Fama-French type models
- **Probability of Default (PD) Modelling:**
  - **Question:** How to efficiently detect probability of default and model it in a non balances dataset?
  - **Method:** Probabilistic regression type models based on selected features

Outlook and Potential Challenges

- **Stock Return Analysis:**
  - Non-stationarity and structural breaks in volatility regimes leading to potential outliars.
  - Non-normality of returns.
- **Credit Card Default Modelling:**
  - **Class Imbalance:** Models risk underpredicting rare defaults without re-weighting or threshold tuning.
  - Potential Multicollinearity in the data and selection of key factors for modelling.

References

Credit Card Default Dataset

- **Paper:** Yeh, I.-C., & Lien, C.-H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473–2480. https://doi.org/10.1016/j.eswa.2007.12.020
- **Repository:** Yeh, I.-C. (2009). *Default of Credit Card Clients* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C55S3H (CC BY 4.0).

Fama-French Factor Data

- **Foundational Paper:** Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56. https://doi.org/10.1016/0304-405X(93)90023-5
- **Data Source:** French, K. R. (2026). *Data Library: Fama/French 3 Factors [Daily]*. Tuck School of Business, Dartmouth College. http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Financial Asset Pricing Data

- **Market Data:** Yahoo! Finance. (2026). *Historical Equity Price Data*. https://finance.yahoo.com/ (Fetched via R-Paket `quantmod`).